

Digital self-comb transition boundaries — `np.diff` waterfalls

Author: rfi-analyst · **Campaign:** marjum-2026-07 · **Rev 2**

What this is

Time-difference waterfalls across eight boundaries of the **digital self-comb** (1.953125 MHz = exactly 8.000 channels, channel-locked, residue 0 mod 8) during its 07-17/18 era.

Rev 2 rebuilds the file selection. Rev 1 chose its files from box-air file-level `lock8`, and that turned out to be the wrong ground truth in two demonstrable ways (below). State and timing now come exclusively from **box-gnd**, per-integration — Aaron's 2026-09-20 methodology decision. Box-gnd is conducted rather than antenna-switched, so it sees through the calibration and VNA cadence.

Box-gnd決定 determines when; it does not narrow what is plotted. Every figure still shows both autos and the cross, as before.

Why the file-level list was replaced

`flags/diag/comb_inventory.jsonl` computes one `lock8` per (file, input) from a median over all 240 integrations. Good for characterising a comb, wrong for dating transitions:

- **It misses short episodes.** `corr_20260717_163125Z.h5` publishes `lock8 = 0.0` on both inputs while carrying the comb for its first 72 integrations. A feature with under roughly half-file duty cycle washes out of a 240-sample median and the file reads clean.
- **Where the antenna is disconnected it dates the antenna returning, not the comb starting.** `corr_20260717_162916Z.h5` 's published detection rests entirely on its final 29 `RFANT` integrations.

Re-derived per-integration on box-gnd, the era contains **22 episodes / 43 transitions**, of which **11 episodes are shorter than one file** and so were structurally invisible before. The aggregate duty cycle is unchanged (87.1% vs the file-level 86%), which is the reassurance that this resolves structure rather than inventing it.

Provenance

- State and timing: `curation/self_comb_transitions_boxgnd.jsonl`, from `flags/diag/self_comb_per_integration_boxgnd.jsonl` (`flagging/scan_self_comb_per_integration.py` + `derive_self_comb_transitions.py`), box-gnd, 8-integration blocks (~4.3 s).
- The statistic is `comb_inventory.tooth_contrast_locked` at spacing 8, **imported from that module, not reimplemented**, so it cannot drift from the detector behind the published inventory.
- Data access: `eigsep_data.index.MetadataIndex` + `eigsep_data.bundle.load_bundle` . `antenna="cross"` resolves the `04` key per file and fixes conjugation, so there is no reason to drop to raw `h5py` .
- Units: raw correlator counts. Frequencies MHz, times UTC.

numpy 2.2.6

The corrected transition list

Read from the product rather than restated here, so the notebook cannot drift from it.

input	0 (box-gnd) -- authoritative for state and timing
block_seconds	4.3
state_rule	ON when lock8>=30.0 and phase==0; OFF when lock8==0.0; else hold
n_episodes	22
duty_cycle	0.8706
caveat	time resolution is one block (~4.3 s); episodes shorter than ~2 integrations can still be missed

	t_start_utc	t_end_utc	duration_s	shorter_than_file
0	15:36:22	15:38:27	129.3	False
1	15:39:10	15:41:53	167.3	False
2	15:43:35	15:49:49	378.3	False
3	15:50:32	16:14:18	1430.3	False
4	16:22:11	16:22:58	51.3	True
5	16:26:41	16:29:50	193.3	False
6	16:34:29	16:36:00	95.3	True
7	16:36:08	16:37:09	65.3	True
8	16:37:21	16:37:56	39.3	True
9	16:38:22	16:39:05	47.3	True
10	16:39:48	16:39:48	4.3	True
11	16:50:06	17:03:12	790.3	False
12	17:03:25	17:03:51	30.3	True
13	17:55:44	17:55:44	4.3	True
14	17:56:31	18:13:25	1018.3	False
15	18:14:47	18:44:08	1765.3	False
16	18:44:21	20:20:39	5782.3	False
17	20:22:22	20:23:48	90.3	True
18	20:24:23	02:51:51	23252.3	False
19	02:52:42	02:52:55	17.3	True
20	02:53:12	02:53:16	8.3	True
21	02:53:34	03:00:08	398.3	False

11 of 22 episodes are shorter than one file, i.e. invisible to a file-level median.

Which transitions land in the selected files

Each selected file is listed with the box-gnd transitions that fall inside it. Note that almost every transition is interior to a file — at 4.3 s resolution there is no reason for a physical switch to align with a 129 s file boundary, and the fact that the old list appeared to put them on file boundaries was an artefact of the file-level median.

	t_utc	kind	file	block
1	15:38:27	ON->off	corr_20260717_153953Z.h5	10
2	15:39:10	off->ON	corr_20260717_153953Z.h5	20
3	15:41:53	ON->off	corr_20260717_154202Z.h5	28
4	15:43:35	off->ON	corr_20260717_154410Z.h5	22
5	15:49:49	ON->off	corr_20260717_155037Z.h5	19
6	15:50:32	off->ON	corr_20260717_155037Z.h5	29
7	16:14:18	ON->off	corr_20260717_161623Z.h5	1
11	16:29:50	ON->off	corr_20260717_163125Z.h5	8
17	16:37:56	ON->off	corr_20260717_164001Z.h5	1
18	16:38:22	off->ON	corr_20260717_164001Z.h5	7
19	16:39:05	ON->off	corr_20260717_164001Z.h5	17
20	16:39:48	off->ON	corr_20260717_164001Z.h5	27
21	16:39:48	ON->off	corr_20260717_164001Z.h5	27
23	17:03:12	ON->off	corr_20260717_170338Z.h5	24
24	17:03:25	off->ON	corr_20260717_170338Z.h5	27

20260717_153953Z ON->off 15:38:27 / off->ON 15:39:10
 20260717_154202Z ON->off 15:41:53
 20260717_154410Z off->ON 15:43:35
 20260717_155037Z ON->off 15:49:49 / off->ON 15:50:32
 20260717_161623Z ON->off 16:14:18
 20260717_163125Z ON->off 16:29:50
 20260717_164001Z ON->off 16:37:56 / off->ON 16:38:22 / ON->off 16:39:05 / off->ON 16:39:48 / ON->off 16:39:48
 20260717_170338Z ON->off 17:03:12 / off->ON 17:03:25

Load

`patterns=` restricts the index to the selected files and `cache=False` keeps the notebook from writing an `.eigsep_index.h5` sidecar into the shared campaign data directory.

```
indexed 1920 integrations from 8 files
skipped: none
--- box-gnd (auto, input 0) ---
1920 rows x 1024 channels (0.000-249.756 MHz)
antenna 'box-gnd' via keys ['0']
8 files used, 0 skipped

--- box-air (auto, input 4) ---
1920 rows x 1024 channels (0.000-249.756 MHz)
antenna 'box-air' via keys ['4']
8 files used, 0 skipped

--- cross (box-gnd x box-air) ---
1920 rows x 1024 channels (0.000-249.756 MHz)
antenna 'cross' via keys ['04']
8 files used, 0 skipped
```

Timebase

All selected files have `sync_consistent = True`, so the index's `time_best` is trustworthy and is used directly. That is not generally true of this campaign — `header/times` is corrupt in 642 of 5,120 files and the standing fallback is the filename close time — so it is checked here rather than assumed.

file	n_int	sync_ok	utc_start	transitions
corr_20260717_153953Z.h5	240	True	15:37:45	ON->off 15:38:27 / off->ON 15:39:10
corr_20260717_154202Z.h5	240	True	15:39:54	ON->off 15:41:53
corr_20260717_154410Z.h5	240	True	15:42:03	off->ON 15:43:35
corr_20260717_155037Z.h5	240	True	15:48:29	ON->off 15:49:49 / off->ON 15:50:32
corr_20260717_161623Z.h5	240	True	16:14:15	ON->off 16:14:18
corr_20260717_163125Z.h5	240	True	16:29:17	ON->off 16:29:50
corr_20260717_164001Z.h5	240	True	16:37:53	ON->off 16:37:56 / off->ON 16:38:22 / ON->off ...
corr_20260717_170338Z.h5	240	True	17:01:30	ON->off 17:03:12 / off->ON 17:03:25

all selected files: sync_consistent = True

int32 accumulator wrap — repaired on the autos only

Auto accumulators are int32 and wrap. Autos are non-negative by construction, so `raw < 0` detects a wrap with no threshold (`detectors.overflow_mask`).

This matters more here than almost anywhere else, because **this notebook differences along time**: a single unrepaired wrap is a 2^{32} step and would set the colour scale on its own.

The cross is deliberately left alone. A cross visibility is genuinely signed in both real and imaginary parts, so `raw < 0` carries no information about wrapping there, and applying the auto repair would corrupt real data.

```
box-gnd wraps repaired: 2811
box-air wraps repaired: 21
cross left as-is (complex, signed): dtype complex128
1024 channels, 0.000-249.756 MHz
```


Differencing

`np.diff` is taken **within each file**, never across a file join. The selected files are not contiguous, so differencing a naively concatenated stack would manufacture a bright artificial row at every boundary. Each file's 240 integrations therefore give 239 difference rows, with a `NaN` row inserted between files as a separator.

For the cross the displayed quantity is ΔV — the signed change in visibility amplitude — so all three panels show the same kind of quantity. (ΔV , which also responds to pure phase rotation, is a different question and is not plotted here.)

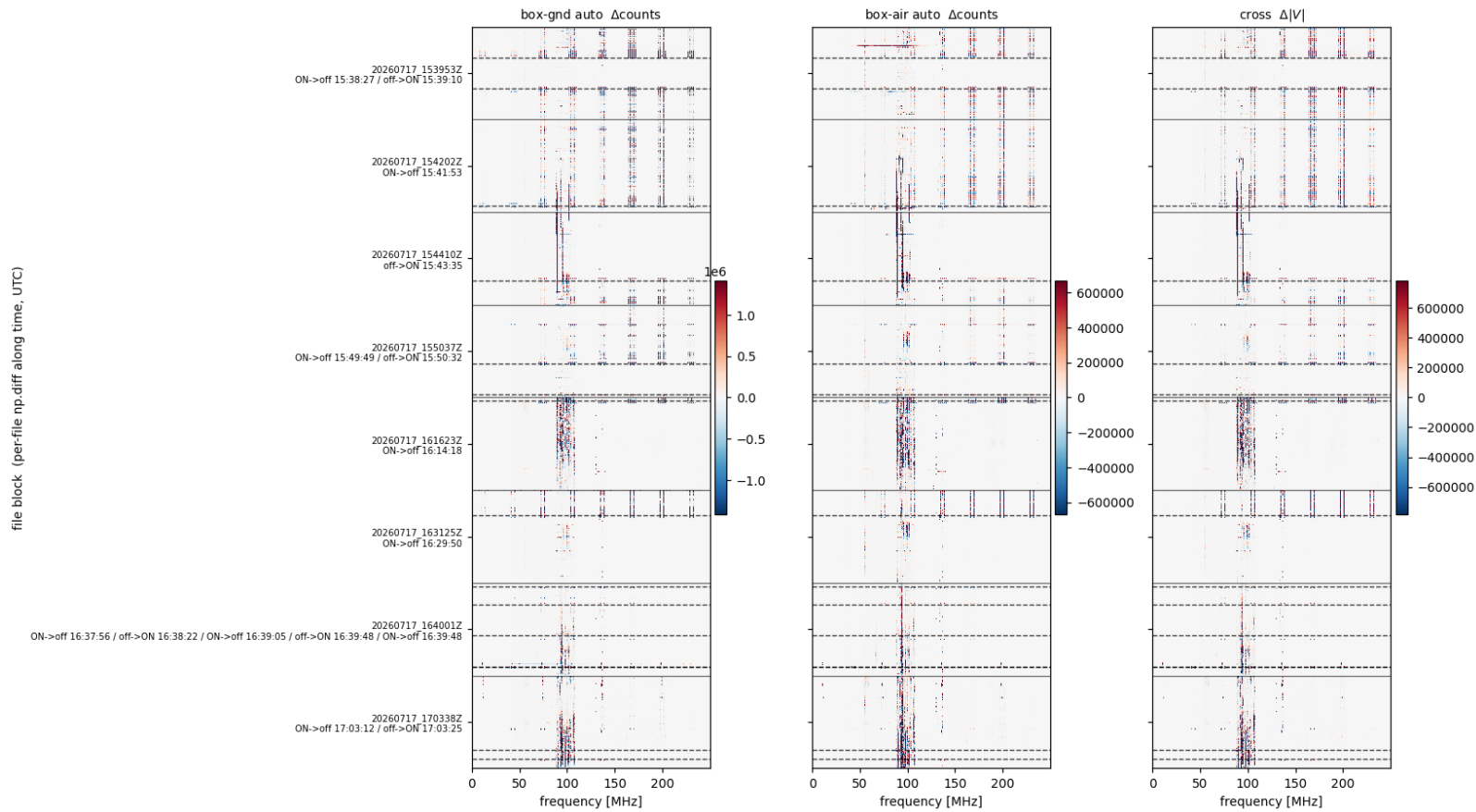
```
stacked diff shape: (1919, 1024)
antenna-to-antenna difference rows: 1912 of 1919 (99.6%)
15 box-gnd transitions fall inside the selected files
```

The figure

Three panels, `sharex=True, sharey=True`: box-gnd auto, box-air auto, cross amplitude. Frequency on x; concatenated per-file difference rows down y, each block labelled with its file and the box-gnd transitions inside it. **Black arrows on the left mark where box-gnd says each transition happens** — that is the ground truth this rev is built on.

The colour scale is symmetric about zero, per panel (the panels differ in absolute scale by orders of magnitude), and set from a robust percentile of antenna-to-antenna rows.

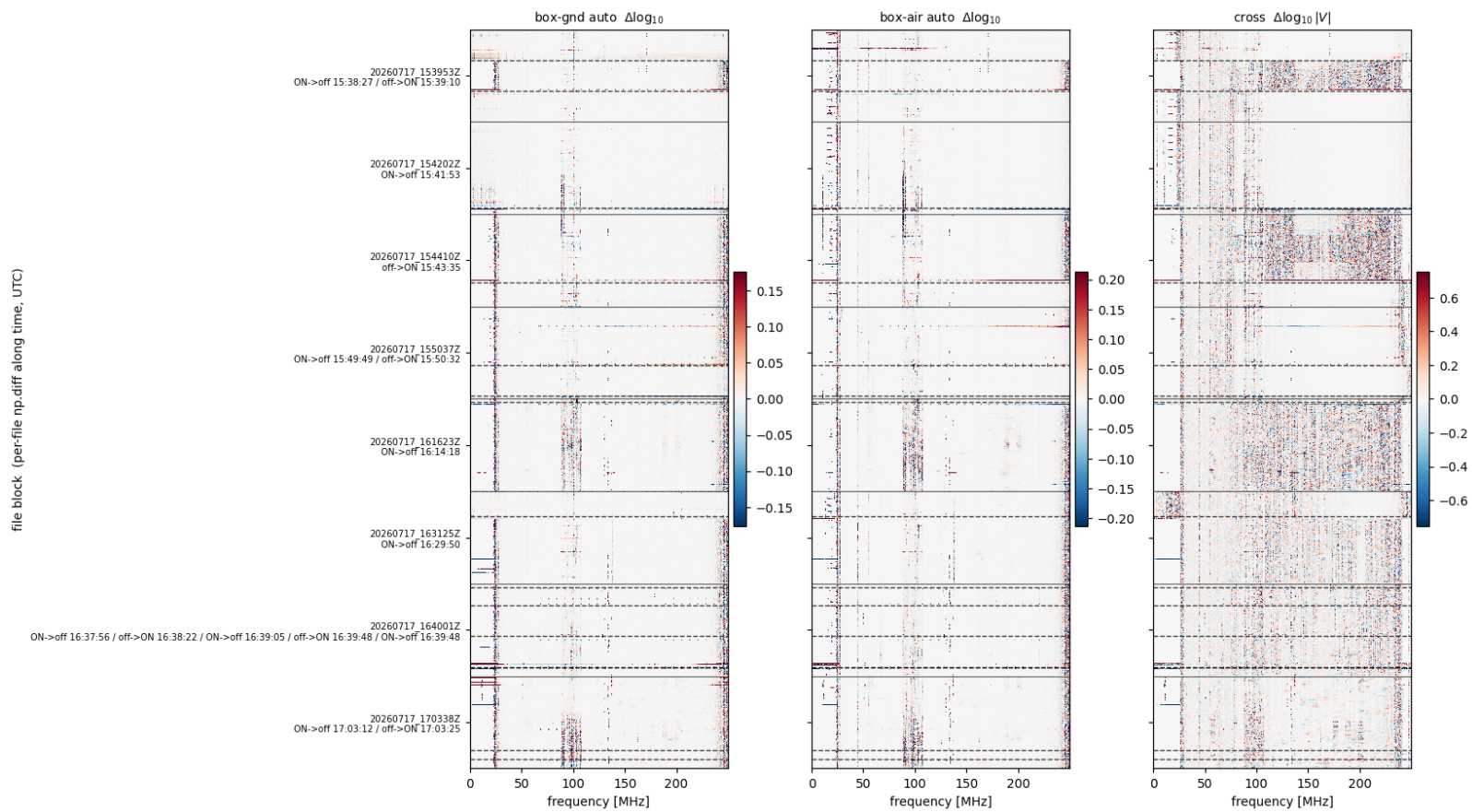
Digital self-comb boundaries — time-difference waterfalls (raw counts)
dashed lines = box-gnd transitions, per-integration ground truth



Same difference, log space

The linear panels are dominated by the brightest channels, which is what a difference of raw counts does. The comb is narrowband and low-amplitude, so the same difference in $\log_{10}$ space — a fractional change per integration — is what makes it legible. Same operation, same data.

Same boundaries, difference taken in log space
(fractional change per integration — narrowband features legible)

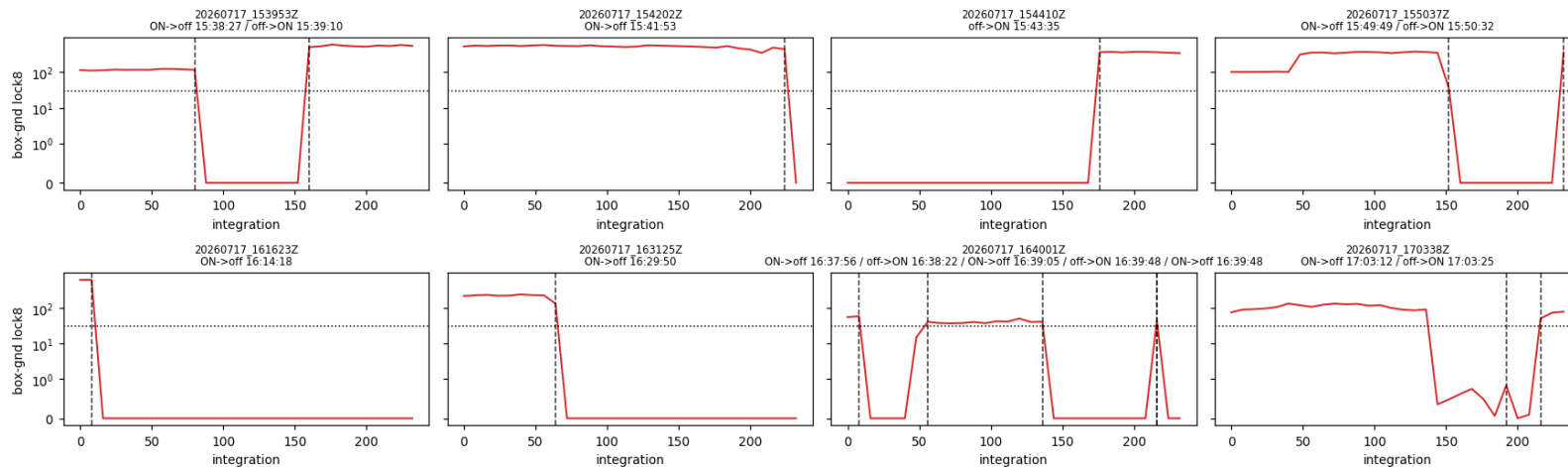


The per-integration state trace

The box-gnd `lock8` trace behind every transition above, for each selected file. This is the evidence for the boundary list, not an illustration of it.

The state rule is hysteretic, and deliberately so. A plain threshold chatters: the transition count runs $73 \rightarrow 197$ as the threshold moves $5 \rightarrow 80$, and 37 of 63 apparent off-runs are under 13 s — the statistic wobbling across a threshold inside a continuously-on stretch. What makes a clean rule possible is the statistic's floor: on comb-free data the log-residual is exactly zero on more than half the in-band channels, the MAD scale collapses, and `tooth_contrast_locked` returns exactly `0.0`. Of 1,380 sub-threshold blocks in the era, **1,176 are exactly 0.0** and only 204 lie in (0, 30). So "off" is a hard value, not a small one, and the rule is: ON at `lock8 >= 30`, OFF at `lock8 == 0.0`, hold in between. That gives 22 episodes, stable at 22–23 for any ON threshold in 20–40.

box-gnd per-integration comb state (8-integration blocks, ~4.3 s)
dotted = ON threshold 30; dashed = derived transition



Six of the eight are unambiguous — 153953Z , 154202Z , 154410Z , 155037Z , 161623Z and 163125Z switch between ~ 0 and 10^2 – 10^3 in a single block, square-wave, nothing near the threshold.

164001Z and 170338Z are the marginal cases and are kept deliberately. Both spend time in the 20–60 range where the hysteresis is doing real work rather than merely formalising an obvious call: 164001Z toggles four times in 129 s, and 170338Z 's last third wanders either side of the threshold. They are the honest worst case for this detector, and a selection containing only the clean six would misrepresent how well it does. If the derived transitions in those two look wrong to you, that is the thing to say — it bears directly on decision 3 below.

The confound, and why it is now gone

Rev 1 had a serious problem. Its six files were picked from box-air file-level lock8 , and in that selection all three ON -> off files were 100% RFANT while all three off -> ON files carried calibration/VNA states — a perfect 3/3 split. "Comb detected" and "calibration cadence running" were not separable, so nothing in the waterfalls could distinguish the comb switching on from the cal hardware switching on.

That confound was **an artefact of the ground truth, not of the comb**. A box-air file-level statistic can only register a change when the antenna happens to be connected, so it preferentially dates transitions to cal-cadence edges — it manufactures the correlation it then appears to show.

With box-gnd per-integration as the state source it disappears:

state	frac_RFANT	RFANT
file		
corr_20260717_153953Z.h5	1.0	240
corr_20260717_154202Z.h5	1.0	240
corr_20260717_154410Z.h5	1.0	240
corr_20260717_155037Z.h5	1.0	240
corr_20260717_161623Z.h5	1.0	240
corr_20260717_163125Z.h5	1.0	240
corr_20260717_164001Z.h5	1.0	240
corr_20260717_170338Z.h5	1.0	240

on-sky fraction by selected file:

```

20260717_153953Z 100.0%
20260717_154202Z 100.0%
20260717_154410Z 100.0%
20260717_155037Z 100.0%
20260717_161623Z 100.0%
20260717_163125Z 100.0%
20260717_164001Z 100.0%
20260717_170338Z 100.0%

```

all eight files pure RFANT: True

All eight selected files are 100% RFANT, and across the whole era **39 of the 43 box-gnd transitions have both sides in pure- RFANT files**. There is no cal/VNA switching anywhere in this selection, so the panels above cannot be showing calibration steps — every feature in them is an antenna-connected, on-sky difference.

This is the substantive payoff of the methodology change, and it also retires rev 1's open question. Rev 1 asked whether its panels showed the comb or the cal cadence, and could not answer from its own data. Under box-gnd ground truth the question does not arise: the transitions are in clean antenna data to begin with.

The era-wide numbers quoted in rev 1 — comb ON 176 pure- RFANT vs 29 cal-active, comb off 49 vs 56 — were computed against the file-level box-air state and should not be carried forward. They describe the old ground truth's behaviour, not the comb's.

Caveats

- **Time resolution is one block, ~4.3 s.** Episodes shorter than about two integrations can still be missed. This is 30× finer than the file-level median it replaces, not a guarantee of completeness.
- **A returned lock8 = 0.0 is ambiguous in principle** — it means "no measurable tooth structure or a degenerate MAD scale". Checked on 750 sampled era blocks: 4.0% had a degenerate scale and every one had an unnormalised best-phase tooth excess of exactly 0.00000, so the degeneracy never hid a comb in this dataset. `tooth_excess_log10` is carried in the trace so this stays auditable.
- **Scope is the documented era**, 07-17 15:37:44 → 07-18 03:00:13. Because the file-level method under-counts short episodes, short episodes outside the era may also have been missed; the era bounds themselves inherit the old method. Not yet checked.
- **Nothing here is an attribution.** No pointing test, no null control, no far-field discriminator. This is morphology and timing only.

Decision requested

1. **Adopt** `curation/self_comb_transitions_boxgnd.jsonl` **as the authoritative self-comb transition list?** It supersedes file-level `lock8` for timing only; `comb_inventory.jsonl` remains correct for characterisation and I have not modified it.
2. **Extend the per-integration scan beyond the era?** Measured cost is ~29 s for 313 era files, so the full campaign is roughly 8 minutes. Given that the file-level method demonstrably misses sub-file episodes, this is the only way to know whether the era bounds are themselves right — and there are already two unexplained box-gnd detections on 07-12 in the file-level inventory.
3. **Do the 22 episodes need a minimum-duration filter for downstream use?** Eleven are shorter than one file and two are a single 4.3 s block. They are real by this detector, but a flag-mask consumer may want them merged or dropped rather than toggling state twice inside one integration window.